Product Requirement Document (PRD): LocalLore

Project Overview

The goal of this project is to develop **LocalLore**, an agentic web application that transforms standard geographical searches into immersive historical and cultural discoveries. Users enter a location using an autocomplete search bar, which immediately opens a dynamic, interactive map centered on that location. An autonomous AI agent then scans, filters, and synthesizes Wikipedia data to unearth interesting stories, historical events, notable figures, and hidden gems nearby.

To keep operational costs at essentially zero, the system uses a high-efficiency caching strategy via Supabase (PostgreSQL with PostGIS), ensuring that previously generated lore maps load instantly without re-invoking the AI. These curated narratives are plotted onto the map as pins, allowing users to move around, click markers, and read engaging "Lore Cards" with headlines and direct links to read more.

Level: Medium
Type of Project: AI Agent Development, Geospatial Web App, Caching Strategy
Skills Required: Next.js (React & TypeScript), Vercel AI SDK, Interactive Mapping APIs (Mapbox), Supabase (PostgreSQL & PostGIS)

Technical Stack Architecture

- **Frontend & Backend Hosting: Next.js (App Router)** deployed on **Vercel (Hobby Tier)**.
- **Maps & Autocomplete: Mapbox Search Box API** (leveraging the 50,000 free monthly search sessions tier) paired with **Mapbox GL JS** for rendering the interactive map canvas.
- **Agent Framework & LLM: Vercel AI SDK** (`streamObject` tool-calling framework) paired with **OpenAI GPT-4o-mini** via a direct pay-as-you-go developer key.
- **Database & Caching Layer: Supabase (PostgreSQL + PostGIS)** to calculate bounding boxes and cache coordinates.
- **Primary Data Source: Wikipedia API** (`geosearch` endpoint).

Key Features & Milestones

Milestone 1: Debounced Search & Instant Map Focus

- **Debounced Autocomplete Search:** A clean search bar with a 300ms debounce buffer to limit unnecessary API requests while typing. It suggests locations seamlessly via Mapbox.
- **Immediate Coordinate Handoff:** Selecting a suggestion immediately grabs the latitude and longitude, hiding the search layout and launching a full-screen interactive map centered on those coordinates.

Milestone 2: Smart Caching Layer (Supabase PostGIS)

- **Geospatial Cache Lookup:** Before calling any AI, the backend queries Supabase using PostGIS to check if a user has searched within a specific radius (e.g., a 2km bounding box) of these coordinates recently.
- **Instant Cache Hit Delivery:** If a match is found in Supabase, the app bypasses the AI completely, serving the saved lore data instantly to the map pins for zero cost.

Milestone 3: The Vercel AI SDK Lore Agent (Cache Miss)

- **Autonomous Wikipedia Tool-Calling:** If the location is a cache miss, the Vercel AI SDK invokes the Wikipedia Geosearch API tool to fetch raw text summaries of nearby geotagged articles.
- **AI Filtering & Schema Enforcement:** The agent uses `streamObject` to enforce a strict JSON schema via Zod, filtering out uninteresting entries and synthesizing the raw summaries into a punchy, 2-3 sentence hook and headline.
- **Cache Hydration:** The fresh JSON response is automatically written to Supabase to cache it for the next user.

Milestone 4: Interactive Map Exploration

- **Dynamic Pin Placement:** Clickable custom markers drop down at the exact coordinates returned by the lore payload.
- **Exploration Canvas:** Users can freely pan and zoom around the map.
- **Lore Cards:** Clicking a marker opens a polished card UI overlay showcasing the snappy headline, the synthesized narrative hook, and a "Read More" button linking out to the source Wikipedia page.

Target Audience & UX Priorities

The app targets curious travelers, history buffs, and locals eager to explore the hidden background of their surroundings. The UI must be lightweight, mobile-responsive for on-the-go mapping, and prioritize sub-second rendering by relying heavily on the database cache.